Anterior triangle of the neck: It encloses four suprahyoid muscles; the digastric, stylohyoid, mylohyoid, geniohyoid.

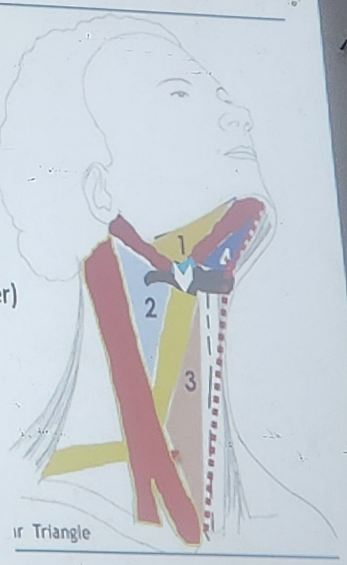
It also encloses four infrahyoid muscles; sternohyoid, sternothyriod, thyrohyoid and omohyoid.

Boundaries of the anterior triangle: **Anteriorly;** The midline of the neck from symphysis menti to supra sternal notch. **Posteriorly;** the anterior boundary of sternocleidomastoid muscle. **Base:** Lower border of mandible and the line extending from angle of mandible to mastoid process. **Apex;** Where the anterior border of sternocleidomastoid muscle meet the anterior midline of the neck.

1. Digastric triangle
2. Carotid triangle
3. Muscular triangle
4. Submental triangle

Fascial sling (investing layer)

Division of anterior triangle



The triangle is divided into four parts by digastric and superior belly of omohyoid into the following:

(i)Digastric (submaxillary) triangle.

(ii) Carotid triangle.

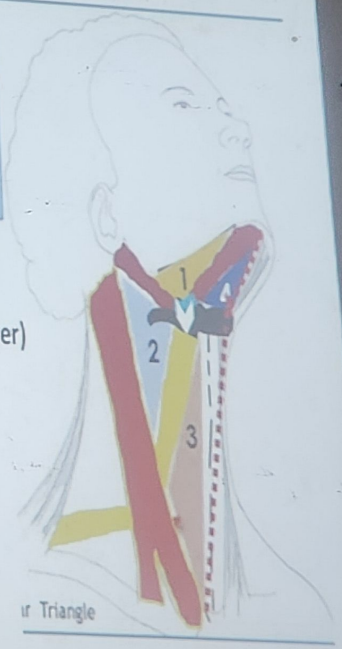
(iii) Muscular triangle.

(iv) Submental triangle.

1. Digastric triangle
2. Carotid triangle
3. Muscular triangle
4. Submental triangle

Fascial sling (investing layer)

Division of anterior triangle



Submental triangle: It is situated between anterior belly of digastric muscles and the raphe demarcating it with its counterpart.

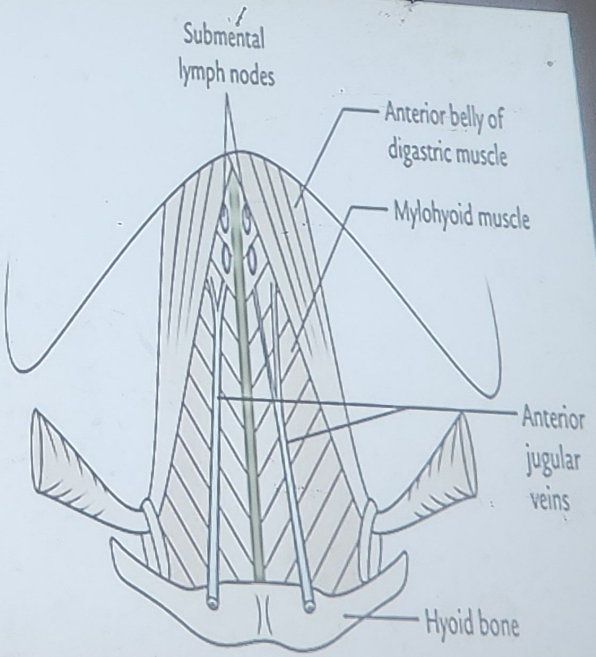
Base: Body of hyoid bone.

Apex: Where the anterior belly of digastric muscle meet the midline raphe at the symphysis menti.

Floor: Mylohyoid muscle.

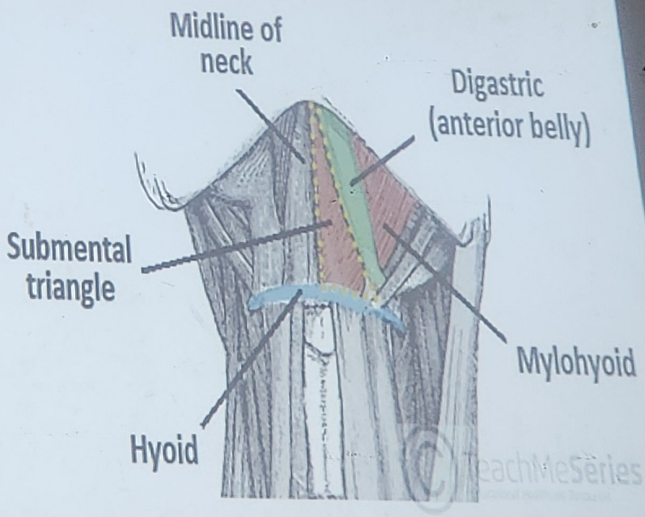
Roof: Deep investing fascia, platysma and skin.

Content: Small submental lymph nodes in superficial fascia between the two anterior belly of digastrics, small submental vein partaking in forming anterior jugular vein.



Clinical conditions:

Diseases of floor of mouth, tip of tongue and incisor teeth cause debilitating enlargement of submental lymph nodes because lymph from these structures is drained into these nodes.



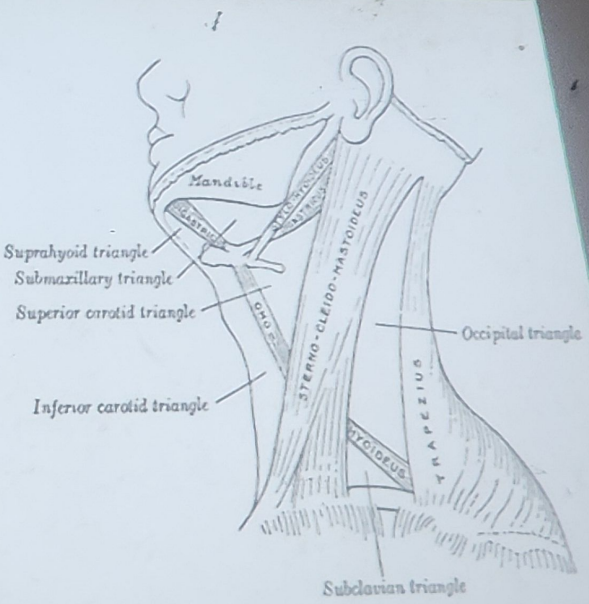
Digastric (submaxillary) triangle: It is an area between the body of mandible and hyoid bone.

Anterior inferior border; Anterior belly of digastric.

Posterior inferior border; Posterior belly of digastric.

Base; Base of the mandible and a line joining the angle of mandible to mastoid process.

Apex; At the tendinous part of digastric muscle attached by pulley to hyoid bone.



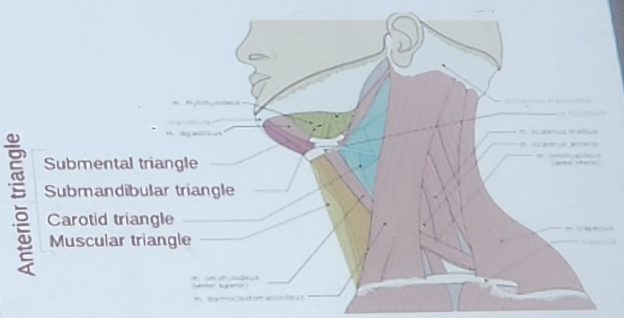
Roof; Skin, superficial fascia, deep fascia.

Floor; anteriorly, mylohyoid, posteriorly, hyoglossus and part of middle constrictor.

Content: Anterior part (structures superficial to mylohyoid), superficial of submandibular gland, submental artery, mylohyoid nerve and vessel, hypoglossal nerve.

Posterior part, lower parotid and external carotid artery.

Deep structures, styloid process and muscular attachments, styloglossus, stylopharyngeus, glossopharyngeus, parotid gland, internal carotid artery and jugular vein.



Carotid triangle: Key triangle in neck.

Anterior superiorly; posterior belly of digastric and stylohyoid.

Anterior inferiorly; superior belly of omohyoid and posteriorly by anterior border of sternocleidomastoid.

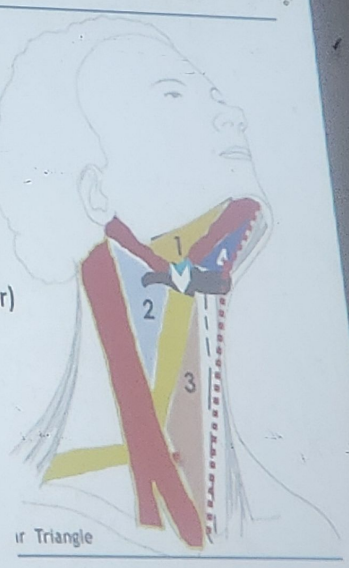
Roof; Skin, platysma, cervical br. of Vii N, transverse cutaneous N of neck, investing deep cervical fascia.

Floor; Middle and inferior constrictor. and thyrohyoid membrane.

1. Digastric triangle
2. Carotid triangle
3. Muscular triangle
4. Submental triangle

Fascial sling (investing layer)

Division of anterior triangle

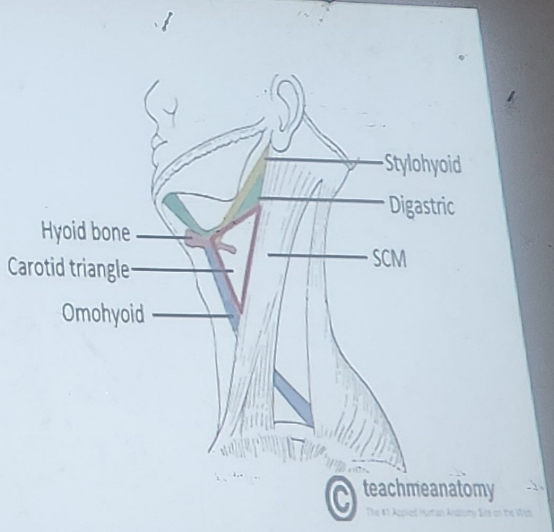


Content: Arterial component; common carotid artery, carotid sinus & body, int. carotid, external carotid arteries.

Venous component; int. jugular vein, lingual, facial and sup thyroid vein.

Nerves; Vagus, sup. laryngeal, hypoglossal, spinal accessory nerve.

Applied anatomy; It is crucial area of clinical procedures and assessment including assessing carotid pulse and carotid endarterectomy. Common complication leads to torrential haemorrhage and death.



Muscular triangle: Anterior median line of neck from hyoid bone to sternum.

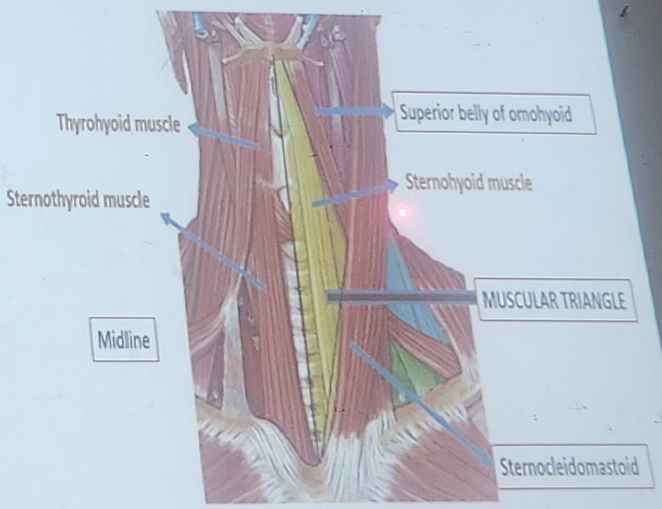
Post sup. Superior belly of omohyoid.

Post inf. Anterior border of sternocleidomastoid.

Content: Infrahyoid muscles.

Applied anatomy; Enlarged submental nodes, goitre, carcinoma of larynx.

Tracheostomy- Opening of trachea and tube inserted to facilitate breathing.



Applied Anatomy:

- The common carotid artery can be compressed against the notable anterior tubercle of transverse process of the 6th cervical vertebra termed carotid tubercle by pressing medially and posteriorly with the thumb. The carotid tubercle of the 6th cervical vertebra is all about 4 cm above the sternoclavicular joint.
- Pressure on carotid sinuses can cause enough slowing of heart rate, fall in blood pressure and cerebral ischemia that will bring about fainting (syncope).
- People with hypersensitive carotid sinuses regularly had unexpected episodes of syncope on rotation of head particularly when wearing a top with tight collar (carotid sinus syndrome).